Dishwashing Quality Improvement

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Business Case

- There has been a measurable percentage of dishes that are still dirty and required re-washing after completion of every cycle of dishwasher. The “dirt” includes greasy residues, food stains, mysterious residue from the dishwasher pod. The re-washing sometimes is done manually or re-loaded in the dishwasher to wash them with the next load.
- Currently on an average about 10.48%(average of 15 loads) of the dishes loaded end up needing re-wash. Out of these at least half of the 5.24% is loaded back in the dishwasher and the other half is manually washed. This not only adds extra cost as more loads are needed than necessary, it also adds to the manual labor of re-working the dishes if not washed correctly.
- By identifying the root causes for the re-loads, I can save on the extra pods used for the extra loads. There is also a considerable amount of time spent on re-washing dishes manually. This will save some time as well by improving the process.

Problem & Goal Statement

- **Y Metric:** % of dishes coming out dirty after one wash
- **Problem**: Currently 10.48% of dishes are coming out dirty out of the dishwasher.
- **Goal**: Reduce the percentage of dishes coming out dirty to 5% by 15-Dec-2025

Operational Definition

- Percentage of defective dishes **is defined as** total percentage of dishes coming out dirty **as measured by** taking the total count of individual dishes including cutlery and lids that has stains or residue on the inner or outer perimeter from both top and bottom rack after every load and converting it as a percentage % defects = No. of. dishes dirty/ Total dishes loaded.

Current State

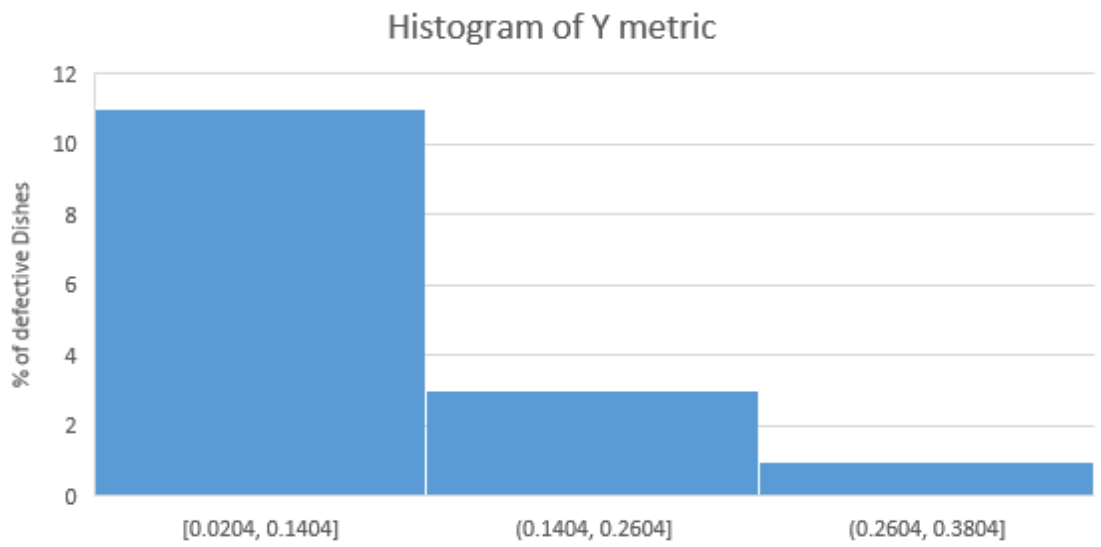
- The current average percentage of defective dishes based on 15 loads done from 30-September to 9-October-2025 is **10.48%**.

Descriptive Statistics	
Mean	0.1048
Standard Error	0.0213
Median	0.0851
Mode	#N/A
Standard Deviation	0.0823
Sample Variance	0.0068
Kurtosis	0.9364
Skewness	1.2353
Range	0.2796
Minimum	0.0204
Maximum	0.3000
Sum	1.5715
Count	15.0000

- The descriptive statistics are not in percentages. We see that the data is right skewed which is evident from the Skewness metric=1.2353,
- Kurtosis is within 1. Hence it is fairly normal

Current State – Normality Tests

K-S and Histogram:



Kolmogrov-Smirnov test for normality	
Summary Stats	
Sample Count	15
Mean	0.10
Standard Deviation	0.08
α	0.05
Kolmogorov-Smirnov Results	
D-Statistic (K-S Statistic)	0.000
Critical Value	0.338
Outcome	Fail to Reject- Treat as Normal
P-Value	1.2732

- The histogram plot shows the distribution of the Y-metric and it seems to be right skewed.
- The Kolmogorov-Smirnov test for normality was done based on the 15 data points and the Outcome was "Failed to Reject, Treat as normal". The Null Hypothesis of this test is H0: Data is normally distributed and alternative hypothesis is HA: Data is not normally distributed.
- Although skewed to the right, the data is good to be used for statistical tests.

Current State – DPMO and Sigma Level

DISCRETE

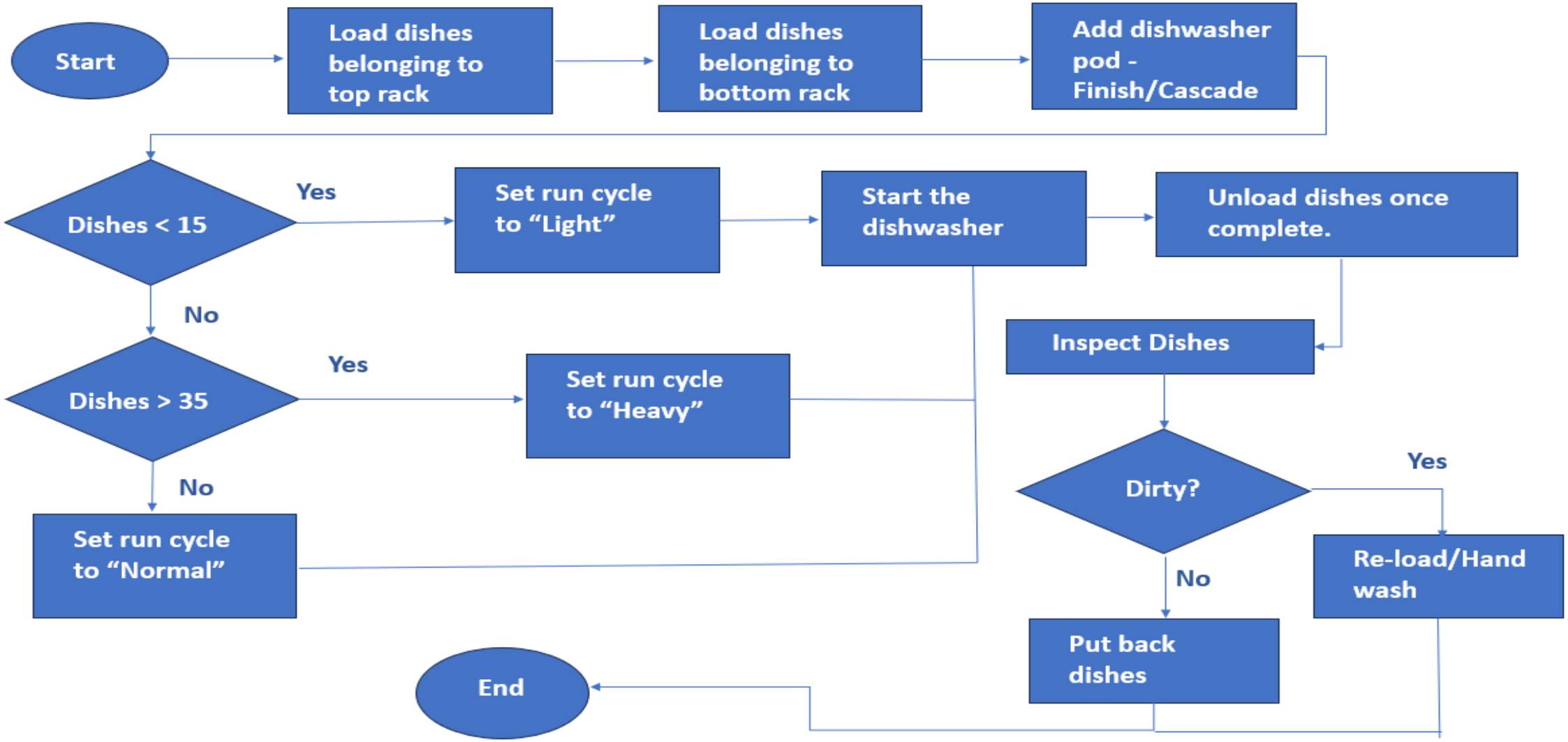
- DPMO and Sigma level if No .of. defects(Discrete) is considered instead of % of defects(Continuous):

total number of defects observed	D =	62
total number of units processed	N =	582
number of defect opportunities per unit	O =	1
Defects Per Opportunity	DPO =	0.10652921
Defects per Million Opportunities	DPMO =	106529.2096
Yield Rate	Yield =	89.34707904
Sigma Level	Sigma Level=	2.745199519

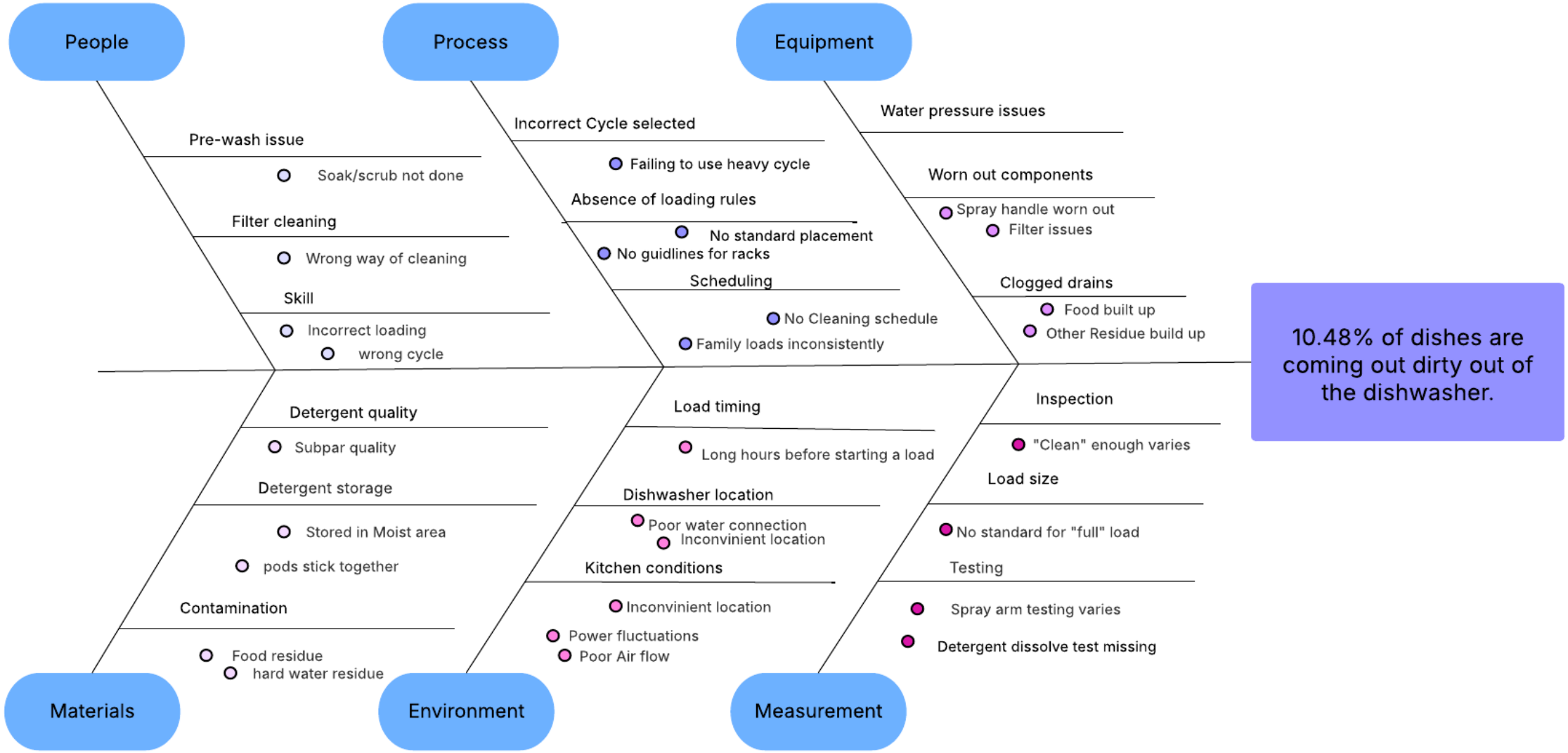
CONTIUOUS

- DPMO and Sigma level if % of defects considered:
- DPMO is not applicable for continuous data.
- Sigma level can be calculated as smaller of the two: Process Mean-LSL/sigma and USL-Process Mean/sigma. 10.48%-LSL/8.23% and USL-10.48%/8.23%

Current State – Process Map – Running Dishwasher



Fishbone Diagram for Dishwasher Quality Improvement



ANALYZE

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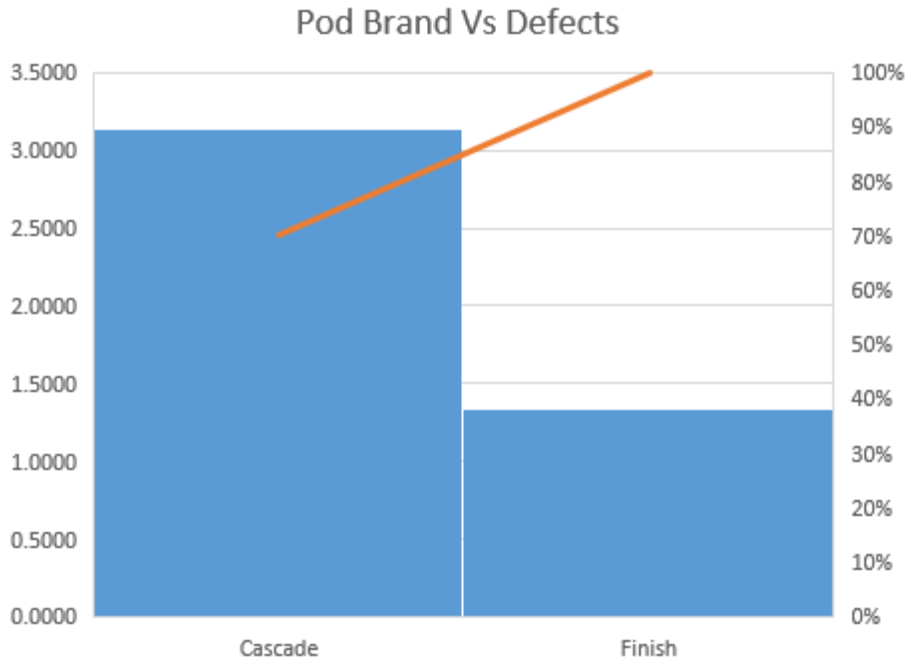
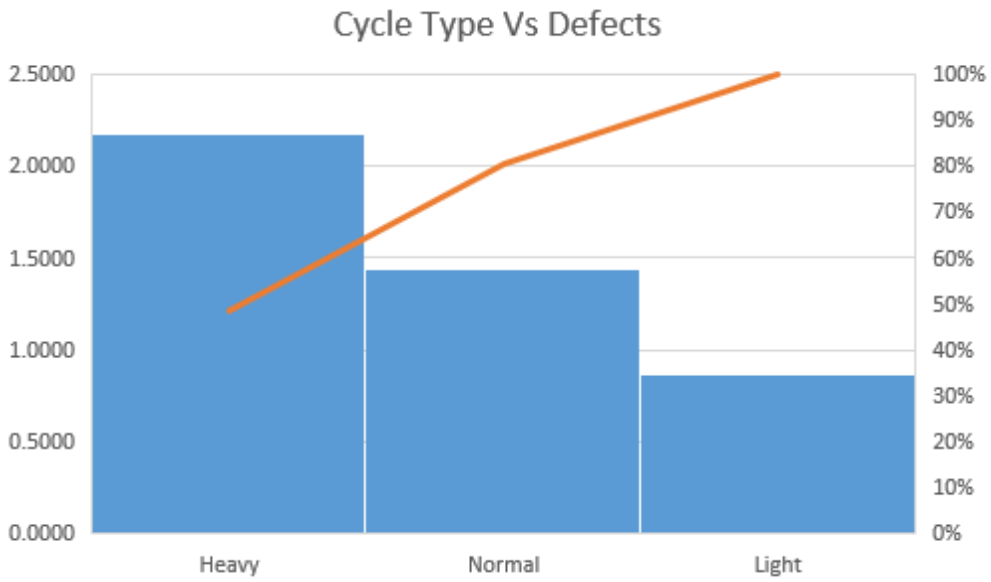
Variables Selected from Fishbone Diagram Analysis

- Number of Dishes in Top Rack(Discrete)
- Number of Dishes in the Bottom Rack(Discrete)
- Load Cycle(Normal/Heavy/Light) – Quantitative(Ordinal)
- Number of dishes Scrubbed
- Number of dishes that are greasy
- Pod type(Cascade/Finish) – Qualitative(Nominal)

Analysis – Pareto Charts

- Which Pod type and wash cycle causes the most percentage of defects?



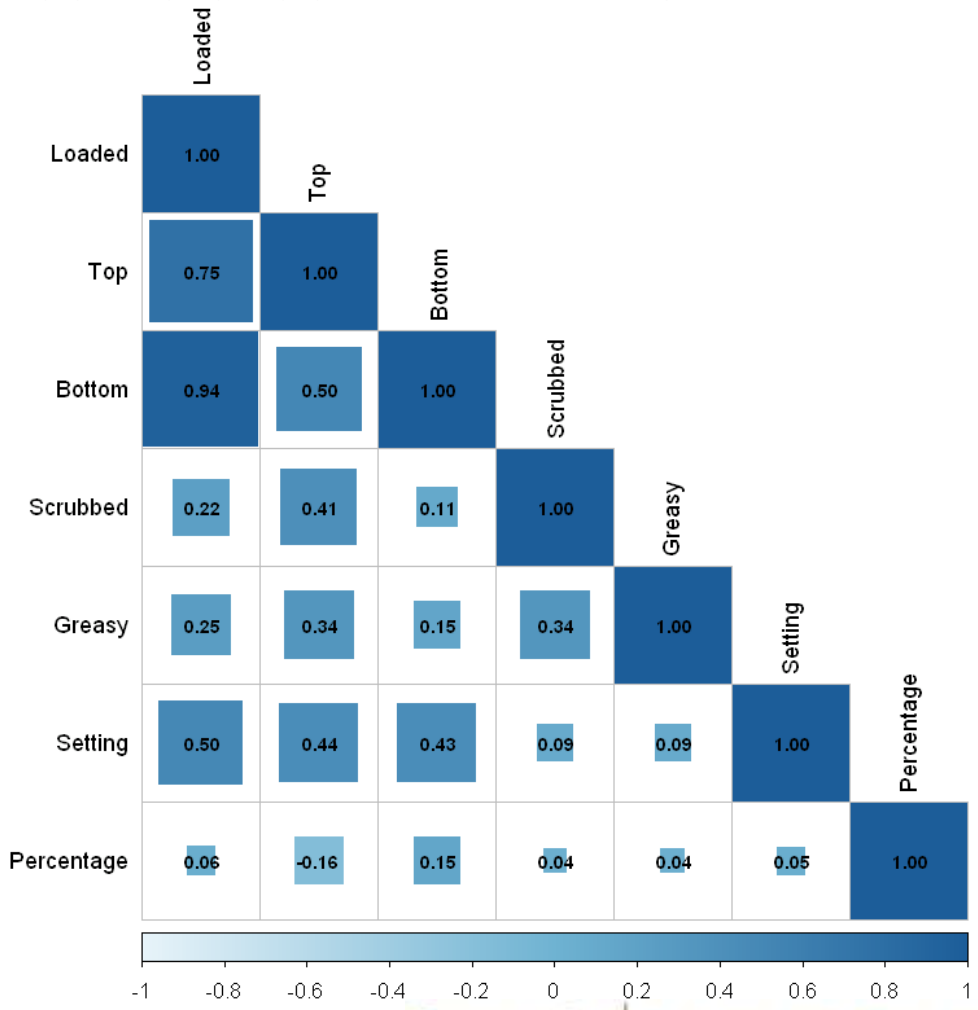


It looks like Cascade Brand has caused the most defects. 80% of the defects have originate from using Cascade Pod.

Surprisingly, more than a light cycle, Heavy and Normal cycle have contributed to 80% of the defects.

Analysis – Correlation Matrix

- A correlation Plot between numerical variables gives us an idea to determine which X variables might be most correlated with Y. The following plot was developed using R.



- Size of the squares represents the strength of correlation.
- Darker squares correspond to positive correlation and lighter squares corresponds to negative correlation.
- There is overall a weak correlation between the Percentage defects and other factors.
- Dishes loaded on the Top rack has a weak negative correlation with percentage of defects.
- The cycle setting doesn't seem to exhibit much correlation with the output.
- The Pod type is not included here as it is categorical.

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Regression Analysis – Includes all factors

- The output percentage of defects variable was regressed onto all X variables and following are the results.

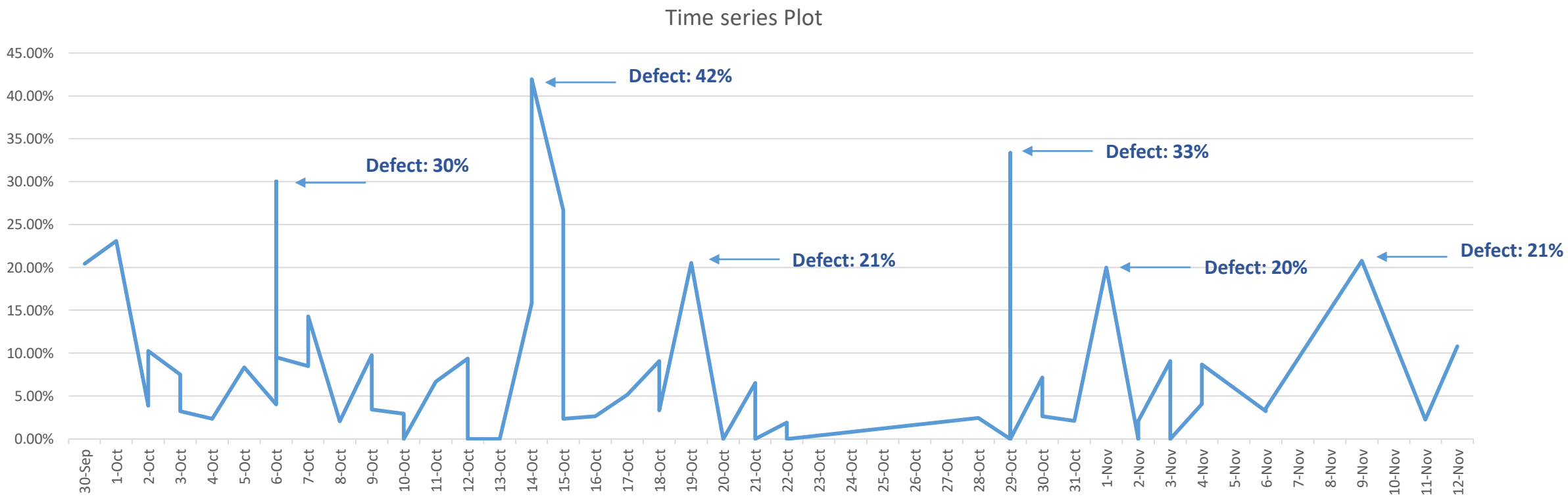
SUMMARY OUTPUT	
Regression Statistics	
Multiple R	0.47355041
R Square	0.224249991
Adjusted R Square	0.111119781
Standard Error	0.088284595
Observations	56

The overall R-squared of using all the X variables to regress onto Y is about 0.5 which is average. It means the given X factors explain 50% of the variability in the output.

	Coefficients	Standard Error	t Stat	P-value
Intercept	0.060921612	0.053574	1.137158	0.261117387
No.of.Dishes Loaded	0.00640957	0.011464	0.559085	0.578703091
Dishes on Top Rack	-0.018813422	0.011943	-1.57523	0.121771639
Dishes on Bottom Rack	-0.003025637	0.011749	-0.25752	0.797874708
No.of.dishes Scrubbed	0.006632152	0.006678	0.993091	0.325645707
No.of.dishes greasy	0.005463488	0.007921	0.689743	0.493677227
Dishwasher Setting	0.01885595	0.020942	0.90037	0.372418398
Pop Brand Encoded	0.057794355	0.024415	2.367198	0.022003065

- The individual coefficients and their statistical significance(p-values) are presented above.
- ‘Dishes on Top rack” and “Pod Brand” are the two variables that have smaller p-value and has statistical significance. These are the main contributors to the variability in the output(defect percentage)

Analysis – Time series Plot

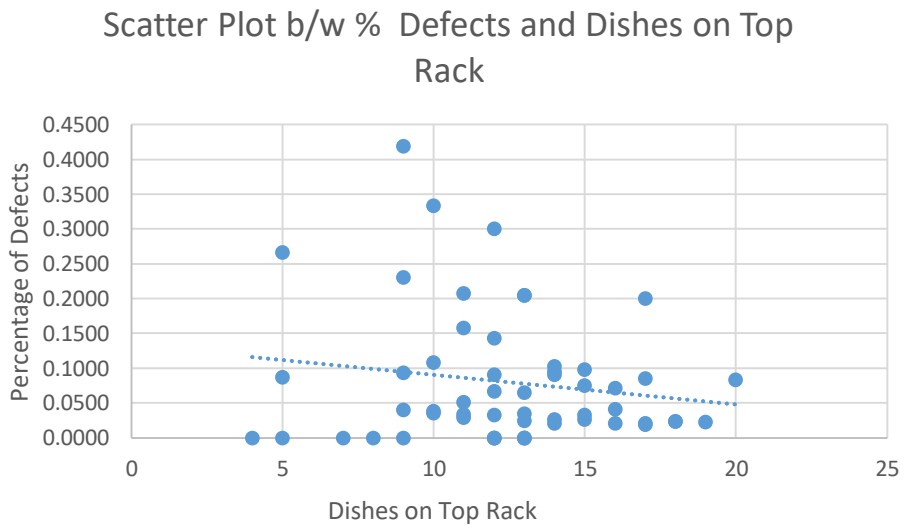


- Unusually large number of defects have occurred on days: 6-Oct,14-Oct,19-Oct,29-Oct,01-Nov and 09-Nov.
- For all 6 runs ,Cascade Pod was used.
- Out of 6 runs, 4 runs were “heavy” setting.
- These runs also had an average of 1-2 dishes that were greasy with an average of 1 dishes scrubbed.

Analysis – Dishes on Top Rack

- A marginal Regression model was build using just the Dishes on Top Rack X variable to see its influence.

	Coefficients	Standard Error	t Stat	P-value
Intercept	0.133020058	0.045306649	2.935994	0.004876448
Dishes on Top Rack	-0.004277156	0.003509877	-1.21861	0.22829101



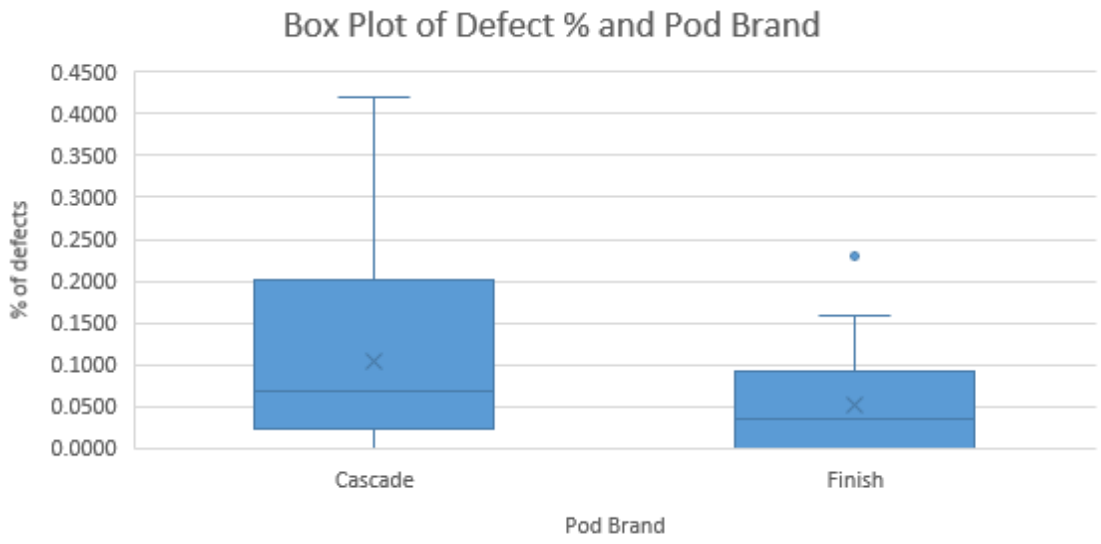
- The marginal regression model does not prove statistical significance for “Dishes on Top Rack” field.
- The Scatter Plot between the Response and ‘Dishes on Top Rack’ variable shows a negative trend although the correlation is minimal. As the number of dishes on Top Rack increased, the defects tend to decrease.

Analysis – Pod Brand

- A marginal Regression model was build using just the Pods Brand X variable to see its influence.

	Coefficients	Standard Error	t Stat	P-value
Intercept	0.051619231	0.017770075	2.90484	0.005315
Pop Brand Encoded	0.052860769	0.024278544	2.177263	0.03385

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Brand	1	0.0389	0.03892	4.74	3.39E-02
Residuals	54	0.4433	0.00821		

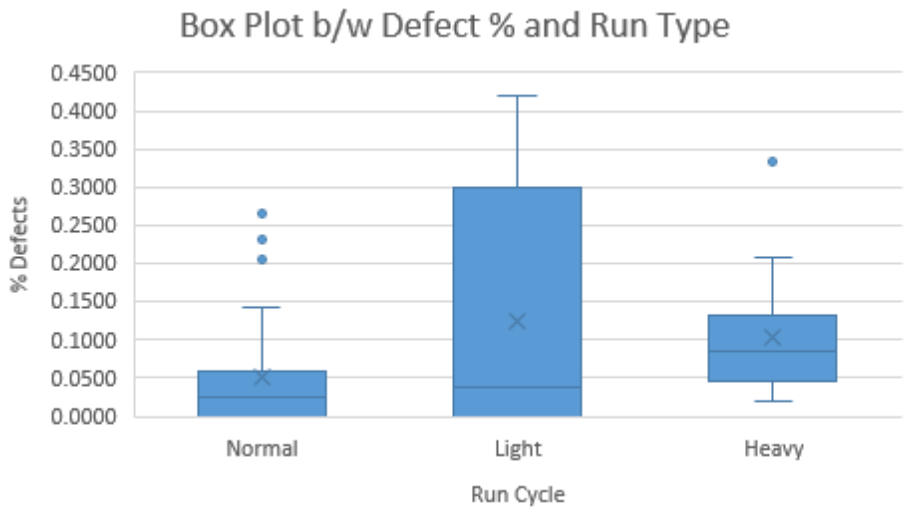


- The marginal regression model shows a low p-value and the variable pod Type is “statistically significant” at level 0.05
- The box plot between the Pod type and percentage of defects clearly shows a difference in the median defects which seems to be higher for Cascade. This however does not provide statistical significance.
- ANOVA equal means test was performed to identify if the means of the two categories of Pods are statistically significantly different. The p-value is small 0.03 making the test statistically significant. Hence, there is difference in means of the percentage of defects between the two pod types.

Analysis – Run Type

- Even though, multiple linear Regression did not prove statistical significance on the dishwasher setting, analysis of variance(ANOVA) was done.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
LoadType	2	0.0479	0.023942	2.921	0.0626
Residuals	53	0.4344	0.008196		

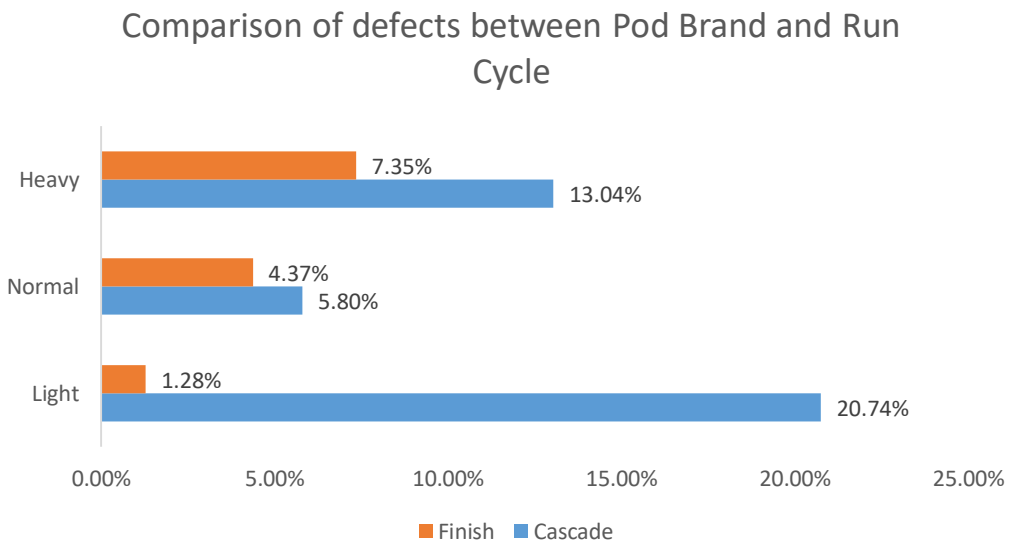


When comparing the defects between the different types of loads(Light , Normal, Heavy) we can see that 'Heavy' type had a lot of defects.

Analysis – Best combination of Run Cycle for the Pod Brand

- A pivot table was designed to find the best Run Cycle for a given pod Type.

Row Labels	Average of Percentage - Y	Interpretation
Cascade	10.45%	Overall Cascade has higher defects
Heavy	13.04%	Poor performance
Light	20.74%	Very poor performance
Normal	5.80%	Good performance
Finish	5.16%	
Heavy	7.35%	Poor performance
Light	1.28%	Very Good performance
Normal	4.37%	Good performance
Grand Total	7.99%	Overall Finish has Lower defects



- We observe that 'Finish' Brand performs best in terms of defects with 5.16% overall average defects.
- Within the Cascade Brand, Light cycle has the worst performance with 21% of defects and Normal Cycle has the best performance with 5.8% defects.
- Within Finish Brand, Light cycle has the least percentage of defects with only about 1.28% defective vessels.

Recommendations

1. Switch to 'Finish' as Primary Pod:

- ANOVA test and Box plot shows a clear winner when it comes to the Brand of pod used. Pareto also revealed that Cascade has caused 80% of the defects. Finish had the lowest defect rate overall ~5.16% and even lower for 'Light' cycle.
- Expected reduction in % defects from ~10.48% to ~5%.
- Cost for Finish per pod is $0.30 and for Cascade is about $0.40 . 'Finish' is beneficial in both ways namely; it saves cost while reducing defects
- **Control:** Create a Purchasing Standard: only Finish pods should be restocked and keep 1 backup box of Cascade only for emergencies.

Recommendations

2. Avoid 'Light' cycle when using Cascade:

- Overall Light cycle has caused a lot of defects, especially for Cascade, the pivot table and stacked bar plot shows that Light cycle using cascade pod has caused about ~21% of defects on an average while with Finish pod the defects are only about 1.28% which is very minimal.
- When there is a need to do a 'Light' load, prefer Finish over Cascade.
- When only Cascade is available, use it with 'Normal' Run cycle setting. The defects % using Normal cycle and Cascade is only about 5.8%.
- Heavy-cascade Runs have caused spikes in defects as demonstrated in the Time Series Plot – Avoid this combination at any cost.
- **Control:** Create a checklist of Run Type describing which Run type is suitable for which pod Brand.

Recommendations

3. Improve Load Distribution (Top vs Bottom Rack Balance):

- Although minimal, the Variable 'Dishes on Top Rack' has a negative correlations with defects which is evident from the overall regression and Marginally regressing this X variable onto Y.
- This means, more load on Top rack has caused lower defects. This could be due to the fact that the Spray arm is present below it and overloading Bottom rack could potentially block the spray arm and overloading top rack doesn't.
- Maintaining a well-balanced load and avoiding overloading at the Bottom and shifting some of the load to Top rack has shown to improve the performance.
- **Control:** Use a Top Rack first loading rule where maximum usage of Top rack needs to be done before moving to the bottom rack.

Appendix A: Data Collection

The same premium variety of cascade and Finish pods were purchased and used alternatively and Data was collected from **30-Sep to 12-Nov** and has **56 samples** in total.

DATA COLLECTION PLAN					
Project (or question to answer):		Reduce Percentage of Dishes coming out dirty from the dishwasher.			
What	Data Type	How Measured	Related Conditions	Sampling Notes	Recording info
% of dishes coming out dirty per wash	Continuous	After every wash, post the drying cycle, count the number of dishes that are still dirty(includes visible stains and/or residue from the dishwasher pod, including spoons and cutlery) and count the total number of dishes(including spoons and cutlery) and convert to a %. (no. dirty dishes ÷ total dishes in load) × 100	Number of dishes loaded, No. of dishes in top rack, No. of Dishes in Bottom rack, No. of greasy dishes loading(visibly greasy), No. of. dishes soaked/scrubbed, Setting of dishwashers – Light, Normal, Heavy, Brand of dishwasher Pod(Finish/Cascade), No.of.hours until dishwasher switched on(Time b/w first dish placed inside until dishwasher is switched on in minutes)	Record every load for 6 weeks or until 50-60 samples collected.	Recorded manually by me,at home, using a spreadsheet

Date	No.of.dishes dirty	No.of.Dishes Loaded	Percentage - Y	Dishes on Top Rack	Dishes on Bottom Rack	No.of.dishes Scrubbed	No.of.dishes greasy	Dishwasher Setting(1/2/3 - Light/Normal/Heavy)	Pod Brand
30-Sep	9	44	20.45%	13	31	2	2	2	Cascade
01-Oct	9	39	23.08%	9	30	1	3	2	Finish
02-Oct	1	26	3.85%	10	16	1	2	1	Finish
02-Oct	4	39	10.26%	14	25	2	3	3	Finish
03-Oct	3	40	7.50%	15	25	3	5	3	Finish
03-Oct	1	31	3.23%	15	24	4	1	2	Cascade
04-Oct	1	43	2.33%	18	25	5	7	1	Cascade
05-Oct	4	48	8.33%	20	28	4	6	3	Cascade